

Week 1 Project

Data Analysis with Vibe Coding:

Mastering Agentic AI Bootcamp | The Gen Academy

What this week is about

This week is focused on helping you learn how to build data applications using vibe coding.

During the live session, we have walked you through how to use tools like Codex (or you can use any AI coding assistant of your choice) to build a Streamlit application for data visualization using a CSV dataset.

The goal is learning how to collaborate with AI coding tools to rapidly prototype useful applications.

At the end of this week, you will have a working data application that analyzes and visualizes a dataset using AI-assisted development workflows.

How this project works

You have two paths. Pick whichever you prefer.

Path A: Build a Stock Portfolio Analyzer	During the session, we demonstrated how to vibe code a Streamlit app using a sample CSV dataset. We've also shared a solution kit. You are free to replicate the exact same project, modify it, use a different dataset, or tweak the prompts to perform different kinds of analysis.
Path B: Build Your Own Vibe-Coded App	You can build any application of your choice using any framework, tooling stack, or AI coding assistant. The goal is simply to explore AI-native coding workflows and ship something functional.

Recommended stack for the live demo We will primarily use Streamlit + CSV datasets during the walkthrough because it is one of the fastest ways to prototype AI-powered data applications. However, you are completely free to use any stack or framework you prefer.

Deliverable

Submit a document where you have added the screenshots of how you went about doing the vibe coding project. If you ended up choosing path B, then explain the problem statement and the entire step-by-step breakdown of how you build your app using vibe coding

Also submit a screen-recording of you showing a working demo of the app, & walking through it

What you will learn

Through this project, you will learn how to:

- Use AI coding assistants for rapid prototyping
- Build interactive data applications with minimal boilerplate
- Work with CSV datasets and visualize insights
- Collaborate with coding agents using structured prompting
- Iterate quickly using vibe coding workflows
- Turn ideas into working demos faster

Suggested Workflow

A typical workflow for this project could look like this:

Step	What you might do
1	Load a CSV dataset into your application
2	Use AI coding tools to generate a Streamlit UI
3	Add charts, filters, tables, and visualizations
4	Improve the UX through iterative prompting
5	Experiment with additional analysis workflows
6	Optionally add AI-powered interactions or summaries

Part 1: Build what was shown in the session

During the session, we will demonstrate multiple prompts that help guide the coding assistant through the application-building process.

You are encouraged to:

- Reuse the same prompts
 - Modify the prompts
 - Experiment with prompt engineering
 - Iterate on outputs
 - Try different coding agents
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1. Download the csv from <link>
 2. Go to Codex app on your desktop
 3. Upload the csv
 4. Use the following prompts, step by step:
 - a. Prompt 1
 - b. Prompt 2
 - c. Prompt 3
 - d. Prompt 4
 - e. Prompt 5

The goal is to become comfortable treating AI coding systems as collaborative development partners.

Part 2 (optional): Build your own app with vibe-coding

You are encouraged to experiment creatively.

Some examples include:

- Sales analytics dashboard
- Startup metrics dashboard
- Financial analysis app
- Fitness or health tracker
- Social media analytics dashboard
- Research paper trend explorer
- Customer feedback analyzer
- AI-generated insights dashboard
- Any application you want to vibe code

How to submit

Deliverables for Week 1

Submit all three of the following by the end of Week 1.

1	Project documentation	Submit a Google Doc explaining what you built. Include: project overview, datasets used, prompts you used during vibe coding, iterations you tried, and any learnings or observations from the workflow.
2	Video demo	Submit a video (5 minutes or less) where you walk through your application, explain what you built, describe how you used AI coding tools, and demonstrate the final result live.
3	Code base	Upload your code assets to Github and share a link in the form below

Project Deadline (16th August - 11:59pm PT)

Fill out the form to make a submission: <https://forms.gle/wmNBQR1DdgTUhYHy5>

- Only people who have submitted within the deadline will be considered for “Builder of the week”
- If you are unable to submit the project by the deadline, you can submit it later in the same link and just shoot us an email to [Tanish \(tanish@thegenacademy.com\)](mailto:tanish@thegenacademy.com)

** Solutions for use-cases

Given below are some solutions that we have put together for the use cases shared in this particular week's suggested use case section (part 1).

We highly encourage you **NOT** to look at this before you get started with your project. We intentionally don't want you to go through this because it will direct your thinking in a particular direction.

We would rather want you to think through the solution and build this yourself, even if it takes you a little bit more time. Only refer to the following documents if you are absolutely stuck and are unable to make progress. Use them as a hint document rather than replicating the following solutions. If you end up replicating the following solutions, you will not be given scores.

#	Use case	Solution Kit
1	Vibe Coding Project	 Week 1 Vibe Coding Project Solution Kit